

# Buy Side: Assignment 4b

## *Financial Engineering*

### *Risk Based Allocation Strategies*

#### Instructions

- **Delivery:** Friday, May 8<sup>th</sup>, 18:00 to [financial.engineering.polimi@gmail.com](mailto:financial.engineering.polimi@gmail.com), with subject “BS: Assignment 4b, Group #”;
- Deliver
  - a **short** pdf document with the results and a discussion of them;
  - the code written in Python using your preferred IDE;
- Fill the gaps where appropriate and correct the errors (in case of incoherences between this document and the code, follow this document);
- Comment (in detail) the code and use explicative variable names;
- Use the market data provided in the file `sx5e_underlyings.csv`.

#### Case study

A junior analyst working for the quantitative team of Polimi Pension Fund is tasked with studying the family of risk-based allocation strategies to complement and diversify the fund strategic asset allocation. In particular, she is asked to analyze portfolios that rely primarily on the correlation and covariance structure of asset returns rather than on expected returns.

The analyst focuses on the Euro Stoxx 50 constituents, performs monthly rebalancing (end of month), and at each rebalance uses the previous two years of data.

All analyses must be repeated for three covariance estimators evaluated side-by-side: the sample covariance matrix, the constant-correlation shrinkage estimator and the single-factor (market) shrinkage estimator (with the equally-weighted return of the investment universe as the market proxy) from the “Buy Side: Assignment 3”.

Assume total return indices in EUR, use simple returns and ignore the risk-free rate.

#### Questions

1. Part I: Equal Risk Contribution (ERC)
  - a. Implement an equal risk contribution portfolio;
  - b. Evaluate the dispersion of the risk contributions and verify that when the covariance matrix is diagonal the ERC solution coincides with the inverse-volatility portfolio;
  - c. Backtest the ERC portfolio across the three covariance estimators. Report and discuss cumulative return, 63d ewm realised volatility (annualised), drawdown and turnover. Comment on the differences across estimators.
2. Part II: Hierarchical Risk Parity (HRP)
  - a. As a preprocessing step for hierarchical clustering, implement a correlation detoning procedure by removing the first principal component from the correlation matrix. Discuss the intuition for this procedure and why removing it should help an agglomerative clustering algorithm. To help you gain insight, visualize and report the dendrogram and the quasi-diagonalized correlation matrix before and after detoning on the last rebalance date.

- b. Implement a generic hierarchical risk parity procedure parametric in:
    - i. The linkage method (focus on single, ward, complete and average);
    - ii. The distance metric used to build the linkage matrix;
    - iii. The volatility allocation traversal:
      1. Recursive bisection: Lopez de Prado's halving of the dendrogram leaf order, regardless of the actual hierarchy;
      2. Dendrogram iteration: at every step split the cluster with the largest merge distance, following the actual hierarchy.
  - c. Set `linkage='ward'` and `distance='euclidean'` and, for each of  $\{\text{sample, constant-correlation, market-factor}\} \times \{\text{detoning on/off}\}$ , compute the Rand index (`sklearn.metrics.rand_score`) between the clusters at consecutive rebalances for  $k = 5, 8, 10$  and 15 clusters. Report the 12m ewm smoothed time series and discuss the following:
    - i. Does correlation detoning improve cluster stability?
    - ii. Does applying shrinkage prior to clustering provide additional benefits?
  - d. Discuss the sensitivity of HRP weights. In particular, run the HRP on  $\{\text{linkage}\} \times \{\text{allocator}\} \times \{\text{estimator}\} \times \{\text{detoning}\}$  and summarise each variant by its mean turnover and its mean Hefindhal concentration index.
    - i. How does the choice of linkage propagate to the weights?
    - ii. How do the two allocation traversals differ? When do they produce similar weights, and when do they differ?
3. Part III: Connecting the Dots
- a. Compute the average L1 distance between the weights produced by the same strategy (choose one representative HRP strategy) under different covariance estimators. Are ERC and HRP equally robust to the choice of estimator?
  - b. Compare the performance of the ERC and HRP portfolios (choose one representative based on the prior analysis) in terms of: cumulative performance, realized volatility (63d ewm annualized), drawdowns and turnover.
  - c. You may observe that, in this exercise, shrinkage techniques do not have the same impact as in *Buy Side: Laboratory 3*. Discuss what this difference reveals about the sensitivity of risk-based allocation techniques compared to optimization-based approaches.